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Semiconductor structure and forming method thereof

Abstract

The present disclosure provides a semiconductor structure and a forming method thereof, including: a gate structure is located on a substrate; a plurality of doped regions, located in the substrate, and located at two sides of the gate structure, the doped region includes a first doped region and a second doped region, a concentration of doped ions in the first doped region is greater than a concentration of doped ions in the second doped region, and the first doped region is far from a sidewall of the gate structure; an electrical contact layer, the electrical contact layer is in contact with a sidewall of the first doped region far from the gate structure, and a top surface of the electrical contact layer is higher than a surface of the substrate; and a dielectric layer, the dielectric layer fills a space between the electrical contact layer and the gate structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present disclosure is a continuation application of International Patent Application No. PCT/CN2022/074554, filed on Jan. 28, 2022, which claims the priority to Chinese Patent Application 202110790428.0, titled “SEMICONDUCTOR STRUCTURE AND FORMING METHOD THEREOF” and filed with the China National Intellectual Property Administration (CNIPA) on Jul. 13, 2021. The entire contents of International Patent Application No. PCT/CN2022/074554 and Chinese Patent Application 202110790428.0 are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to, but is not limited to, a semiconductor structure and a forming method thereof.

BACKGROUND

(2) As a semiconductor memory device commonly used in a computer, a dynamic random access memory (DRAM) includes many repeated memory cells. Each memory cell usually includes a capacitor and a transistor. The transistor is provided with a gate connected to a word line, a drain connected to a bit line, and a source connected to the capacitor. A voltage signal on the word line can control on or off of the transistor, and then data information stored in the capacitor is read through the bit line, or the data information is written into the capacitor through the bit line for storage.

(3) With the rapid development of integrated circuit technologies, a density of devices in an integrated circuit is increasingly high, and the critical dimension of a semiconductor device is continuously decreased. Particularly, an effective gate length is shortened. Consequently, a short-channel effect leads to a leakage problem of a source/drain region, posing a challenge to the reliability of the devices.

SUMMARY

(4) The present disclosure provides a semiconductor structure and a forming method thereof.

(5) A first aspect of the present disclosure provides a semiconductor structure, including: a substrate; a gate structure, where the gate structure is located on the substrate; a plurality of doped regions, located in the substrate, and located at two sides of the gate structure, where the doped region includes a first doped region and a second doped region, a concentration of doped ions in the first doped region is greater than a concentration of doped ions in the second doped region, and the first doped region is far from a sidewall of the gate structure; an electrical contact layer, where the electrical contact layer is in contact with a sidewall of the first doped region far from the gate structure, and a top surface of the electrical contact layer is higher than a surface of the substrate; and a dielectric layer, where the dielectric layer fills a space between the electrical contact layer and the gate structure.

(6) A second aspect of the present disclosure provides a forming method of a semiconductor structure, including: providing a substrate and a gate structure, where the gate structure is located on the substrate; performing ion implantation on the substrate to form a plurality of doped regions in the substrate, where the doped regions are located at two sides of the gate structure, where the doped region includes a first doped region and a second doped region, a concentration of doped ions in the first doped region is greater than a concentration of doped ions in the second doped region, and the first doped region is far from a sidewall of the gate structure; removing part of the substrate, to form a trench, where the trench exposes a sidewall of the first doped region far from the gate structure; forming an electrical contact layer, where the electrical contact layer fills up the trench and is in contact with the sidewall of the first doped region far from the gate structure, and a top surface of the electrical contact layer is higher than a surface of the substrate; and forming a

dielectric layer, where the dielectric layer fills a space between the electrical contact layer and the gate structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings incorporated into the specification and constituting a part of the specification illustrate the embodiments of the present disclosure, and are used together with the description to explain the principles of the embodiments of the present disclosure. In these accompanying drawings, similar reference numerals represent similar elements. The accompanying drawings in the following description illustrate some rather than all of the embodiments of the present disclosure. Those skilled in the art may obtain other accompanying drawings based on these accompanying drawings without creative efforts.

(2) FIG. **1** is a schematic structural diagram of a semiconductor structure;

(3) FIG. **2** is a schematic structural diagram of a semiconductor structure according to an embodiment of the present disclosure;

(4) FIG. **3** is another schematic structural diagram of a semiconductor structure according to an embodiment of the present disclosure;

(5) FIG. **4** is still another schematic structural diagram of a semiconductor structure according to an embodiment of the present disclosure;

(6) FIG. **5** is a schematic structural top view of a semiconductor structure according to an embodiment of the present disclosure;

(7) FIG. **6** is another schematic structural top view of a semiconductor structure according to an embodiment of the present disclosure;

(8) FIG. **7** is a schematic structural diagram of a substrate and a gate structure that are provided in a forming method of a semiconductor structure according to another embodiment of the present disclosure;

(9) FIG. **8** is a schematic structural diagram obtained after a dielectric layer is formed in the structure shown in FIG. **7**;

(10) FIG. **9** is a schematic structural diagram obtained after a trench is formed in the structure shown in FIG. **8**;

(11) FIG. **10** is a schematic structural diagram obtained after an electrical contact layer is formed in the structure shown in FIG. **9**;

(12) FIG. **11** is a schematic structural diagram obtained after an initial electrical contact layer is formed in a process of forming an electrical contact layer and a dielectric layer according to another embodiment of the present disclosure;

(13) FIG. **12** is a schematic structural diagram obtained after part of an initial electrical contact layer located on a sidewall of a gate structure is removed in the structure shown in FIG. **11**; and

(14) FIG. **13** is a schematic structural diagram obtained after a groove is filled to form a second dielectric layer in the structure shown in FIG. **12**.

DETAILED DESCRIPTION

(15) The technical solutions in the embodiments of the present disclosure are described below clearly and completely with reference to the accompanying drawings in the embodiments of the present disclosure. Apparently, the described embodiments are merely some rather than all of the embodiments of the present disclosure. All other embodiments obtained by those skilled in the art based on the embodiments of the present disclosure without creative efforts should fall within the protection scope of the present disclosure. It should be noted that the embodiments in the present disclosure and features in the embodiments may be combined with each other in a non-conflicting manner.

(16) As mentioned in the background, the reliability of the semiconductor structure in the prior art is insufficient.
(17) FIG. **1** is a schematic structural diagram of a semiconductor structure.
(18) Referring to FIG. **1**, a semiconductor structure is provided, including: a substrate **100**; a gate structure **120**, where the gate structure **120** is located on the substrate **100**, and the gate structure **120** includes a gate oxide layer **121** and a gate **122**; a plurality of doped regions **110**, located in the substrate **100**, and located at two sides of the gate structure **120**, where the doped region **110** includes a first doped region **111**, a second doped region **112**, and a third doped region **113**, a concentration of doped ions in the first doped region **111** is greater than a concentration of doped ions in the second doped region **112** and a concentration of doped ions in the third doped region **113**, the second doped region **112** is formed in the third doped region **113**, the first doped region **111** is formed in the second doped region **112**, and surfaces of the first doped region **111**, the second doped region **112**, and the third doped region **113** that are close to the substrate **100** are flush with each other; and an electrical contact layer **130**, where the electrical contact layer **130** is located on an upper surface of the doped region **110** and is in contact with the first doped region **111**.
(19) The doped region **110** may be used as a source or drain of the semiconductor structure. It can be learned from the foregoing that, the doped region **110** has gradient doping distribution, and a doping concentration in a region closer to the surface of the substrate **100** is higher, such that a region with a higher doping concentration is slightly far from the gate structure **120**. However, a distance between the gate structure **120** and the region with a higher doping concentration in the source/drain region is still relatively small, and when the gate structure **120** is turned on, the gate structure **120** generates an enhanced electric field in the region, where the enhanced electric field still affects doped ions in the doped region, making the doped region **110** leak a current under the influence of a strong electric field. This manner of only making concentrations of doped ions in the doped region **110** present simple gradient distribution to improve device performance does not meet a requirement of further reduction in a device size, and a current leakage phenomenon in the source/drain region is still serious. In addition, since the subsequently formed electrical contact layer **130** is located on the upper surface of the doped region **110** and is relatively close to the gate structure **120**, the gate structure **120** and the electrical contact layer **130** are also easily short circuited.
(20) An embodiment of the present disclosure provides a semiconductor structure, where a concentration of doped ions in a first doped region is greater than a concentration of doped ions in a second doped region, and the first doped region is far from a sidewall of a gate structure, to ensure that most of the doped ions are far from the gate structure. When the gate structure is turned on, the gate structure generates an enhanced electric field in the region. Because the most of the doped ions are far from the gate structure, the enhanced electric field does not affect doped ions in a source/drain region. This is beneficial to avoiding a current leakage risk of the doped region caused by the influence of a strong electric field, and improving performance of the semiconductor structure. In addition, because the electrical contact layer is in contact with a sidewall of the first doped region far from the gate structure, a distance between the electrical contact layer and the gate structure is relatively long. This effectively avoids short circuit caused by a case in which the gate structure is in contact with the electrical contact layer, and is beneficial to improving the reliability of the semiconductor structure.
(21) FIG. **2** is a schematic structural diagram of a semiconductor structure according to an embodiment of the present disclosure.
(22) Referring to FIG. **2**, a semiconductor structure provided in this embodiment includes: a substrate **200**; a gate structure **220**, where the gate structure **220** is located on the substrate **200**; a plurality of doped regions **210**, located in the substrate **200**, and located at two sides of the gate structure **220**, where the doped region **210** includes a first doped region **211** and a second doped

region **212**, a concentration of doped ions in the first doped region **211** is greater than a concentration of doped ions in the second doped region **212**, and the first doped region **211** is far from a sidewall of the gate structure **220**; an electrical contact layer **230**, where the electrical contact layer **230** is in contact with a sidewall of the first doped region **211** far from the gate structure **220**, and a top surface of the electrical contact layer **230** is higher than a surface of the substrate **200**; and a dielectric layer **240**, where the dielectric layer **240** fills a space between the electrical contact layer **230** and the gate structure **220**.

(23) The substrate **200** is made of a semiconductor material. In this embodiment, the material of the substrate **200** is silicon. In another embodiment, the substrate may be alternatively a germanium substrate, a germanium-silicon substrate, a silicon carbide substrate, or a silicon-on-insulator substrate.

(24) In this embodiment, the gate structure **220** includes a gate oxide layer **221**, a first gate conductive layer **222**, a gate barrier layer **223**, and a second gate conductive layer **224** that are sequentially stacked.

(25) The gate oxide layer **221** is made of an insulating material, for example, silicon dioxide, silicon carbide, or silicon nitride, and is configured to isolate the gate conductive layer from the doped region **210**.

(26) In this embodiment, the first gate conductive layer **222** has a relatively low resistance, and a material thereof may be polycrystalline silicon, used to reduce a contact resistance. The gate barrier layer **223** is configured to bar mutual diffusion of the first gate conductive layer **222** and the second gate conductive layer **224**, is further configured to increase adhesion between the first gate conductive layer **222** and the second gate conductive layer **224**, and a material thereof may be titanium nitride or tantalum nitride. A material of the second gate conductive layer **224** may be tungsten. In another embodiment, the material of the second gate conductive layer may be alternatively another metal material, for example, gold or silver.

(27) The gate structure **220** further includes an isolation structure **225**, and the isolation structure **225** is located on a surface of the gate oxide layer **221**, a surface of the first gate conductive layer **222**, a surface of the gate barrier layer **223**, and a surface of the second gate conductive layer **224**.

(28) In this embodiment, the isolation structure **225** is a single-layer structure, where a material of the isolation structure **225** is silicon nitride; and is mainly configured to perform isolation and insulation. In another embodiment, the isolation structure may be alternatively a multi-layer structure. The isolation structure **225** has relatively high hardness and a relatively large density and can improve an isolation effect, to avoid a case in which the gate structure **220** is electrically connected to another subsequently formed conductive structure, such that occurrence of problems such as short circuit or current leakage is avoided. In addition, the isolation structure **225** has a relatively good corrosion resistance capability, and therefore can avoid damage during a cleaning process.

(29) The doped region **210** may be an N-type doped region or a P-type doped region. In this embodiment, the doped region **210** is an N-type doped region, the doped region **210** is doped with N-type ions, and the substrate **200** is doped with P-type ions. In another embodiment, the doped region is a P-type doped region, the doped region is doped with P-type ions, and the substrate is doped with N-type ions.

(30) The doped region **210** located on one side of the gate structure **220** is used as a source, and the doped region **210** located on the other side of the gate structure **220** is used as a drain.

(31) In this embodiment, the doped region **210** includes the first doped region **211** and the second doped region **212**. A concentration of doped ions in the first doped region **211** is greater than a concentration of doped ions in the second doped region **212**, and the first doped region **211** is far from a sidewall of the gate structure **220**, to ensure that most of the doped ions are far from the gate structure **220**. When the gate structure **220** is turned on, the gate structure **220** generates an enhanced electric field in the region. Because the most of the doped ions are far from the gate

structure **220**, the enhanced electric field does not affect doped ions in the doped region **210**. This is beneficial to avoiding a current leakage risk of the doped region **210** caused by the influence of a strong electric field, and improving performance of the semiconductor structure.
(32) In this embodiment, the doped region **210** further includes a third doped region **213**, and a concentration of doped ions in the third doped region **213** is less than the concentration of the doped ions in the second doped region **212**.
(33) The first doped region **211** is formed in the second doped region **212**, and a sidewall of the second doped region **212** far from the gate structure **220** is flush with the sidewall of the first doped region **211** far from the gate structure **220**. It may be understood that, the second doped region **212** is formed in the third doped region **213**, and a sidewall of the third doped region **213** far from the gate structure **220** is flush with the sidewall of the second doped region **212** far from the gate structure **220**.
(34) When the doped region **210** with this structure is formed, on the basis of ensuring that the most of the doped ions in the doped region **210** are far from the gate structure **220**, the third doped region **213** with a lower doping concentration may be first formed in the entire doped region **210**, and then doping is performed on preset regions for a plurality of times, to form a doped region with a higher doping concentration. In the process of forming the entire doped region **210**, there is no need to worry about a problem of mutual contamination between different doped regions.
(35) FIG. **3** is another schematic structural diagram of a semiconductor structure according to an embodiment of the present disclosure.
(36) Referring to FIG. **3**, in another embodiment, a second doped region **212** and a first doped region **211** are sequentially arranged along a direction far from a sidewall of a gate structure **220**. It may be understood that a third doped region **213** is located on a side of the second doped region **212** far from the first doped region **211**. Through such arrangement, a distance between a sidewall of the first doped region **211** with a maximum doping concentration close to the gate structure **220** and the gate structure **220** is relatively long. This further avoids a current leakage problem of the doped region **210**.
(37) Still referring to FIG. **2**, a material of the electrical contact layer **230** is polycrystalline silicon doped with tungsten metal and the like.
(38) Because the electrical contact layer **230** is in contact with the sidewall of the first doped region **211** far from the gate structure **220**, a distance between the electrical contact layer **230** and the gate structure **220** is relatively long. In this way, the dielectric layer **240** subsequently formed between the gate structure **220** and the electrical contact layer **230** has a relatively large thickness. This can effectively avoid short circuit caused by a case in which the gate structure **220** is in contact with the electrical contact layer **230**, and is beneficial to improving the reliability of the semiconductor structure.
(39) The electrical contact layer **230** includes a first electrical contact layer **231** and a second electrical contact layer **232**, where the first electrical contact layer **231** is in contact with the sidewall of the first doped region **211** far from the gate structure **220**, and the second electrical contact layer **232** is located on the first electrical contact layer **231**. A distance between the second electrical contact layer **232** and the gate structure **220** is greater than a distance between the first electrical contact layer **231** and the gate structure **220** in a direction perpendicular to the sidewall of the gate structure **220**.
(40) It can be learned that the second electrical contact layer **232** in the electrical contact layer **230** faces the gate structure **220**. In addition, the distance between the second electrical contact layer **232** and the gate structure **220** is greater than the distance between the first electrical contact layer **231** and the gate structure **220**. This ensures that the electrical contact layer **230** facing the gate structure **220** has a relatively long distance from the gate structure **220**, and is beneficial to avoiding a short circuit risk between the gate structure **220** and the electrical contact layer **230**.
(41) In this embodiment, a top surface of the first electrical contact layer **231** is not higher than the

surface of the substrate **200**; a sidewall of the second electrical contact layer **232** far from the gate structure **220** is flush with a sidewall of the first electrical contact layer **231** far from the gate structure **220**, and a width of the second electrical contact layer **232** is shorter than a width of the first electrical contact layer **231** in the direction perpendicular to the sidewall of the gate structure **220**.

(42) In this way, a side of the electrical contact layer **230** far from the gate structure **220** is a flat surface. This simplifies a topography of the semiconductor structure, and maximizes the distance between the electrical contact layer **230** and the gate structure **220** on this basis.

(43) In this embodiment, a material of the first electrical contact layer **231** is the same as a material of the second electrical contact layer **232**. That the material of the first electrical contact layer **231** is the same as the material of the second electrical contact layer **232** is beneficial to simplifying the step of forming the entire electrical contact layer **230**. After an entire initial electrical contact layer is formed, the initial electrical contact layer is etched, to remove part of the initial electrical contact layer, such that the first electrical contact layer **231** and the second electrical contact layer **232** are formed.

(44) In another embodiment, a material of the first electrical contact layer and a material of the second electrical contact layer may be alternatively different. For example, the material of the first electrical contact layer is doped polycrystalline silicon, and the material of the second electrical contact layer is tungsten metal. Because the first electrical contact layer is in contact with the doped region, a contact resistance between the doped polycrystalline silicon and the doped region is relatively small. This is beneficial to improving a conductive effect between the electrical contact layer and the doped region.

(45) FIG. **4** is still another schematic structural diagram of a semiconductor structure according to an embodiment of the present disclosure.

(46) Referring to FIG. **4**, in another embodiment, a width of an electrical contact layer **230** remains unchanged in a direction perpendicular to a surface of a substrate **200**. That is, the electrical contact layer **230** is integral. Through such arrangement, a structure of the electrical contact layer **230** is simple while ensuring that a specific distance exists between the electrical contact layer **230** and the gate structure **220**. This simplifies a forming process.

(47) Still referring to FIG. **2**, in this embodiment, the first electrical contact layer **231** is in contact with the entire sidewall of the first doped region **211** far from the gate structure **220**. In another embodiment, the first electrical contact layer may be alternatively in contact with part of the sidewall of the first doped region far from the gate structure.

(48) In this way, a relatively large contact area exists between the electrical contact layer **230** and the doped region **210**. A larger contact area indicates a smaller contact resistance between the electrical contact layer **230** and the doped region **210**. This is beneficial to improving a conductive effect between the electrical contact layer **230** and the doped region **210**, and further improving performance of the semiconductor structure.

(49) FIG. **5** is a schematic structural top view of a semiconductor structure according to an embodiment of the present disclosure.

(50) Referring to FIG. **5**, in this embodiment, a length of a second electrical contact layer **232** is equal to a length of a first electrical contact layer **231** (referring to FIG. **2**) in an extension direction of a sidewall in a doped region **210** (referring to FIG. **2**). In this way, two ends of the first electrical contact layer **231** and two ends of the second electrical contact layer **232** are aligned, and topography of the entire electrical contact layer **230** (referring to FIG. **2**) is relatively flat. This simplifies a forming step.

(51) FIG. **6** is another schematic structural top view of a semiconductor structure according to an embodiment of the present disclosure.

(52) Referring to FIG. **6**, in another embodiment, a length of a second electrical contact layer **232** is shorter than a length of a first electrical contact layer **231** in an extension direction of a sidewall of

a doped region **210** (referring to FIG. **2**).
(53) Because the first electrical contact layer **231** is in contact with a first doped region **211**, the length of the second electrical contact layer **232** is relatively small, and a material of the entire electrical contact layer **230** is reduced while ensuring that a contact area between the entire electrical contact layer **230** and the first doped region **211** remains unchanged. This is beneficial to reducing a volume of the entire semiconductor structure.
(54) In this embodiment, the electrical contact layer **230** may further include a metal semiconductor compound layer, a resistivity of a material of the metal semiconductor compound layer is greater than a resistivity of a material of the first doped region **211**, and the metal semiconductor compound layer is in contact with a sidewall of the first doped region **211** far from the gate structure **220**.
(55) The material of the metal semiconductor compound layer may be titanium nitride, silicon nitride, or the like; the metal semiconductor compound layer is configured to bar mutual diffusion between the electrical contact layer **230** and the doped region **210** and is further configured to increase adhesion between the electrical contact layer **230** and the doped region **210**.
(56) Still referring to FIG. **2**, in this embodiment, a material of the dielectric layer **240** may be an insulating material, for example, silicon oxide, silicon carbide, or silicon nitride.
(57) The material of the dielectric layer **240** is the insulating material. Therefore, the dielectric layer **240** located between the gate structure **220** and the electrical contact layer **230** can effectively avoid a case in which the gate structure **220** is electrically connected to the electrical contact layer **230**.
(58) In this embodiment, the dielectric layer **240** includes a first dielectric layer **241** and a second dielectric layer **242** that are sequentially arranged along a direction far from the sidewall of the gate structure **220**; a sidewall of the first dielectric layer **241** far from the gate structure **220** is flush with the sidewall of the first doped region **211** far from the gate structure **220**; and the second dielectric layer **242** is located between the first dielectric layer **241** and the second electrical contact layer **232**.
(59) That the sidewall of the first dielectric layer **241** far from the gate structure **220** is flush with the sidewall of the first doped region **211** far from the gate structure **220** means that the first dielectric layer **241** covers an upper surface of the doped region **210**. This can ensure that the doped region **210** is not affected when the second electrical contact layer **232** is formed.
(60) In this embodiment, a concentration of doped ions in the first doped region **211** is greater than a concentration of doped ions in the second doped region **212**, and the first doped region **211** is far from the sidewall of the gate structure **220**, to ensure that most of the doped ions are far from the gate structure **220**. When the gate structure **220** is turned on, the gate structure **220** generates an enhanced electric field in the region. Because the most of the doped ions are far from the gate structure **220**, the enhanced electric field does not affect doped ions in the doped region **210**. This is beneficial to avoiding a current leakage risk of the doped region **210** caused by the influence of a strong electric field, and improving performance of the semiconductor structure. In addition, because the electrical contact layer **230** is in contact with the sidewall of the first doped region **211** far from the gate structure **220**, a distance between the electrical contact layer **230** and the gate structure **220** is relatively long. This effectively avoids short circuit caused by a case in which the gate structure **220** is in contact with the electrical contact layer **230**, and is beneficial to improving the reliability of the semiconductor structure.
(61) Another embodiment of the present disclosure provides a forming method of a semiconductor structure. The forming method of a semiconductor structure may be used to form the semiconductor structure provided by the foregoing embodiment. The forming method of a semiconductor structure provided by the another embodiment of the present disclosure is described in detail below with reference to the accompanying drawings.
(62) FIG. **7** to FIG. **13** are schematic structural diagrams corresponding to various steps of a

forming method of a semiconductor structure according to another embodiment of the present disclosure.
(63) Referring to FIG. **7**, a substrate **300** and a gate structure **320** are provided, where the gate structure **320** is located on the substrate **300**.
(64) The substrate **300** is made of a semiconductor material. In this embodiment, the material of the substrate **300** is silicon. In another embodiment, the substrate may be alternatively a germanium substrate, a germanium-silicon substrate, a silicon carbide substrate, or a silicon-on-insulator substrate.
(65) In this embodiment, the gate structure **320** includes a gate oxide layer **321**, a first gate conductive layer **322**, a gate barrier layer **323**, and a second gate conductive layer **324** that are sequentially stacked.
(66) A material of the gate oxide layer **321** is silicon oxide or a high dielectric material, where the high dielectric material may include one or more of materials HfO.sub.2, HfSiO, HfSiON, HfAlO, HfZrO, Al.sub.2O.sub.3, TaO.sub.2, and the like. The high dielectric material is a material whose relative dielectric constant is greater than a relative dielectric constant of silicon oxide, that is, a high-k material.
(67) In this embodiment, the first gate conductive layer **322** has a relatively low resistance, and a material thereof may be polycrystalline silicon. The gate barrier layer **323** is configured to bar mutual diffusion of the first gate conductive layer **322** and the second gate conductive layer **324**, is further configured to increase adhesion between the first gate conductive layer **322** and the second gate conductive layer **324**, and a material thereof may be titanium nitride or tantalum nitride. A material of the second gate conductive layer **324** may be tungsten. In another embodiment, the material of the first gate conductive layer may be alternatively another metal material, for example, gold or silver.
(68) The gate structure **320** further includes an isolation structure **325**, and the isolation structure **325** is located on a surface of the gate oxide layer **321**, a surface of the first gate conductive layer **322**, a surface of the gate barrier layer **323**, and a surface of the second gate conductive layer **324**.
(69) In this embodiment, the isolation structure **325** is a single-layer structure, where a material of the isolation structure **325** is silicon nitride; and is mainly configured to perform isolation and insulation. In another embodiment, the isolation structure may be alternatively a multi-layer structure. The isolation structure **325** has relatively high hardness and a relatively large density and can improve an isolation effect, to avoid a case in which the gate structure **320** is electrically connected to another subsequently formed conductive structure, such that occurrence of problems such as short circuit or current leakage is avoided. In addition, the isolation structure **325** has a relatively good corrosion resistance capability, and therefore can avoid damage during a cleaning process.
(70) Ion implantation is performed on the substrate **300** to form a plurality of doped regions **310** in the substrate **300**, where the doped regions **310** are located at two sides of the gate structure **320**; the doped region **310** includes a first doped region **311** and a second doped region **312**, where a concentration of doped ions in the first doped region **311** is greater than a concentration of doped ions in the second doped region **312**; and the first doped region **311** is far from a sidewall of the gate structure **320**.
(71) The doped region **310** may be an N-type doped region or a P-type doped region. In this embodiment, the doped region **310** is an N-type doped region, the doped region **310** is doped with N-type ions, and the substrate **300** is doped with P-type ions. In another embodiment, the doped region is a P-type doped region, the doped region is doped with P-type ions, and the substrate is doped with N-type ions.
(72) The doped region **310** located on one side of the gate structure **320** is used as a source, and the doped region **310** located on the other side of the gate structure **320** is used as a drain.
(73) In this embodiment, the doped region **310** includes the first doped region **311** and the second

doped region **312**. The concentration of the doped ions in the first doped region **311** is greater than the concentration of the doped ions in the second doped region **312**, and the first doped region **311** is far from the sidewall of the gate structure **320**, to ensure that most of the doped ions are far from the gate structure **320**. When the gate structure **320** is turned on, the gate structure **320** generates an enhanced electric field in the region. Because the most of the doped ions are far from the gate structure **320**, the enhanced electric field does not affect doped ions in the doped region **310**. This is beneficial to avoiding a current leakage risk of the doped region **310** caused by the influence of a strong electric field, and improving performance of the semiconductor structure.
(74) In this embodiment, the formed doped region **310** further includes a third doped region **313**, and a concentration of doped ions in the third doped region **313** is less than the concentration of the doped ions in the second doped region **312**.
(75) The first doped region **311** is formed in the second doped region **312**, and a sidewall of the second doped region **312** far from the gate structure **320** is flush with the sidewall of the first doped region **311** far from the gate structure **320**. It may be understood that, the second doped region **312** is formed in the third doped region **313**, and a sidewall of the third doped region **313** far from the gate structure **320** is flush with the sidewall of the second doped region **312** far from the gate structure **320**.
(76) When the doped region **310** with this structure is formed, on the basis of ensuring that the most of the doped ions in the doped region **310** are far from the gate structure **320**, the third doped region **313** with a lower doping concentration may be first formed in the entire doped region **310**, and then doping is performed on preset regions for a plurality of times, to form a doped region with a higher doping concentration. In the process of forming the entire doped region **310**, there is no need to worry about a problem of mutual contamination between different doped regions.
(77) An initial dielectric layer **343** is formed, where the initial dielectric layer **343** covers a surface of the gate structure **320** and a surface of the substrate **300**, and a thickness of the initial dielectric layer **343** located on the sidewall of the gate structure **320** is equal to a width of a top surface of the doped region **310** in a direction perpendicular to the sidewall of the gate structure **320**.
(78) The thickness of the initial dielectric layer **343** located on the sidewall of the gate structure **320** is equal to the width of the top surface of the doped region **310**, such that a case in which an etching process affects the doped region **310** can be avoided when part of the initial dielectric layer **343** is subsequently removed.
(79) In this embodiment, the initial dielectric layer **343** is formed by using an atomic deposition process. In another embodiment, the initial dielectric layer may be alternatively formed by using a chemical vapor deposition process.
(80) A mask layer **350** is formed on an upper surface of the initial dielectric layer **343** located on an upper surface of the gate structure **320**, where the mask layer **350** is a mask used to subsequently remove part of the substrate **300**.
(81) Referring to FIG. **8**, a dielectric layer **340** is formed, where the dielectric layer **340** fills a space between a subsequently formed electrical contact layer and the gate structure **320**.
(82) In this embodiment, forming the dielectric layer **340** may be: etching part of the initial dielectric layer **343** (referring to FIG. **7**), to expose the surface of the substrate **300**, where the remaining part of the initial dielectric layer **343** is used as the dielectric layer **340**.
(83) In this embodiment, the part of the initial dielectric layer **343** is removed by using a wet etching process.
(84) A sidewall of the formed dielectric layer **340** far from the gate structure **320** is flush with the sidewall of the first doped region **311** far from the gate structure **320**. In this way, when the substrate **300** is subsequently etched by using the dielectric layer **340** and the mask layer **350** as a mask, a formed trench exactly exposes the sidewall of the first doped region **311** far from the gate structure **320**.
(85) Referring to FIG. **9**, part of the substrate **300** is removed, to form a trench **360**, where the

trench **360** exposes the sidewall of the first doped region **311** far from the gate structure **320**.
(86) In this embodiment, the part of the substrate **300** is removed by using the wet etching process, to form the trench **360**. In another embodiment, a dry etching process may be alternatively used. The trench **360** exposes the sidewall of the first doped region **311**, and a subsequently formed electrical contact layer filling the trench **360** may be in contact with a region with a maximum concentration in the doped region **310**. This is beneficial to improving contact performance.
(87) Referring to FIG. **10**, an electrical contact layer **330** is formed, where the electrical contact layer **330** fills up the trench **360** (referring to FIG. **9**) and is in contact with the sidewall of the first doped region **311** far from the gate structure **320**, and a top surface of the electrical contact layer **330** is higher than the surface of the substrate **300**.
(88) In this embodiment, the formed electrical contact layer **330** may be: The electrical contact layer **330** fills the trench **360** and is located on the sidewall of the dielectric layer **340**, and a width of the electrical contact layer **330** in a direction perpendicular to the surface of the substrate **300** remains the same.
(89) Through such arrangement, a structure of the electrical contact layer **330** is simple while ensuring that a specific distance exists between the electrical contact layer **330** and the gate structure **320**. This simplifies a forming process.
(90) In another example, the step of forming the electrical contact layer **330** and the step of forming the dielectric layer **340** include:
(91) Referring to FIG. **11**, a first dielectric layer **341** is formed on the surface of the gate structure **320** after the gate structure **320** is formed, where a sidewall of the first dielectric layer **341** far from the gate structure **320** is flush with the sidewall of the first doped region **311** far from the gate structure **320**; an initial electrical contact layer **333** is formed after the trench **360** (referring to FIG. **9**) is formed, where the initial electrical contact layer **333** fills up the trench **360**, and the initial electrical contact layer **333** is located on a side of the gate structure **320**.
(92) Referring to FIG. **12**, part of the initial electrical contact layer **333** (referring to FIG. **11**) located on the sidewall of the gate structure **320** is removed, the remaining part of the initial electrical contact layer **333** is used as the electrical contact layer **330**, and a sidewall of the electrical contact layer **330** and the sidewall of the gate structure **320** enclose a groove **370**.
(93) The electrical contact layer **330** includes a first electrical contact layer **331** and a second electrical contact layer **332**, where the first electrical contact layer **331** is in contact with the sidewall of the first doped region **311** far from the gate structure **320**, and the second electrical contact layer **332** is located on the first electrical contact layer **331**. A distance between the second electrical contact layer **332** and the gate structure **320** is greater than a distance between the first electrical contact layer **331** and the gate structure **320** in a direction perpendicular to the sidewall of the gate structure **320**.
(94) It can be learned that the second electrical contact layer **332** in the electrical contact layer **330** faces the gate structure **320**. In addition, the distance between the second electrical contact layer **332** and the gate structure **320** is greater than the distance between the first electrical contact layer **331** and the gate structure **320**. This ensures that the electrical contact layer **330** facing the gate structure **320** has a relatively long distance from the gate structure **320**, and is beneficial to avoiding a short circuit risk between the gate structure **220** and the electrical contact layer **330**.
(95) Referring to FIG. **13**, the groove **370** (referring to FIG. **12**) is filled, to form a second dielectric layer **342**, where the first dielectric layer **341** and the second dielectric layer **342** constitute the dielectric layer **340**.
(96) That the sidewall of the first dielectric layer **341** far from the gate structure **320** is flush with the sidewall of the first doped region **311** far from the gate structure **320** means that the first dielectric layer **341** covers an upper surface of the doped region **310**. This can ensure that the doped region **310** is not affected when the second electrical contact layer **332** is formed.
(97) In this embodiment, the semiconductor structure is formed, where a concentration of doped

ions in the first doped region **311** is greater than a concentration of doped ions in the second doped region **312**, and the first doped region **311** is far from the sidewall of the gate structure **320**, to ensure that most of the doped ions are far from the gate structure **320**. When the gate structure **320** is turned on, the gate structure **320** generates an enhanced electric field in the region. Because the most of the doped ions are far from the gate structure **320**, the enhanced electric field does not affect doped ions in the doped region **310**. This is beneficial to avoiding a current leakage risk of the doped region **310** caused by the influence of a strong electric field, and improving performance of the semiconductor structure. In addition, because the electrical contact layer **330** is in contact with the sidewall of the first doped region **311** far from the gate structure **320**, a distance between the electrical contact layer **330** and the gate structure **320** is relatively long. This effectively avoids short circuit caused by a case in which the gate structure **320** is in contact with the electrical contact layer **330**, and is beneficial to improving the reliability of the semiconductor structure.

(98) The embodiments or implementations of this specification are described in a progressive manner, and each embodiment focuses on differences from other embodiments. The same or similar parts between the embodiments may refer to each other.

(99) In the description of this specification, the description with reference to terms such as “an embodiment”, “an exemplary embodiment”, “some implementations”, “a schematic implementation”, and “an example” means that the specific feature, structure, material, or characteristic described in combination with the implementation(s) or example(s) is included in at least one implementation or example of the present disclosure.

(100) In this specification, the schematic expression of the above terms does not necessarily refer to the same implementation or example. Moreover, the described specific feature, structure, material or characteristic may be combined in an appropriate manner in any one or more implementations or examples.

(101) It should be noted that in the description of the present disclosure, the terms such as “center”, “top”, “bottom”, “left”, “right”, “vertical”, “horizontal”, “inner” and “outer” indicate the orientation or position relationships based on the accompanying drawings. These terms are merely intended to facilitate description of the present disclosure and simplify the description, rather than to indicate or imply that the mentioned apparatus or element must have a specific orientation and must be constructed and operated in a specific orientation. Therefore, these terms should not be construed as a limitation to the present disclosure.

(102) It can be understood that the terms such as “first” and “second” used in the present disclosure can be used to describe various structures, but these structures are not limited by these terms. Instead, these terms are merely intended to distinguish one structure from another.

(103) The same elements in one or more accompanying drawings are denoted by similar reference numerals. For the sake of clarity, various parts in the accompanying drawings are not drawn to scale. In addition, some well-known parts may not be shown. For the sake of brevity, a structure obtained by implementing a plurality of steps may be shown in one figure. In order to understand the present disclosure more clearly, many specific details of the present disclosure, such as the structure, material, size, processing process, and technology of the device, are described below. However, as those skilled in the art can understand, the present disclosure may not be implemented according to these specific details.

(104) Finally, it should be noted that the above embodiments are merely intended to explain the technical solutions of the present disclosure, rather than to limit the present disclosure. Although the present disclosure is described in detail with reference to the above embodiments, those skilled in the art should understand that they may still modify the technical solutions described in the above embodiments, or make equivalent substitutions of some or all of the technical features recorded therein, without deviating the essence of the corresponding technical solutions from the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. A semiconductor structure, comprising: a substrate; a gate structure, wherein the gate structure is located on the substrate; a plurality of doped regions, located in the substrate, and located at two sides of the gate structure, wherein one of the plurality of doped regions comprises a first doped region and a second doped region, a concentration of doped ions in the first doped region is greater than a concentration of doped ions in the second doped region, and the first doped region is farther from a sidewall of the gate structure than the second doped region; an electrical contact layer, wherein the electrical contact layer is in contact with a sidewall of the first doped region far from the gate structure, and a top surface of the electrical contact layer is higher than a surface of the substrate; and a dielectric layer, wherein the dielectric layer fills a space between the electrical contact layer and the gate structure.
2. The semiconductor structure according to claim 1, wherein the electrical contact layer comprises a first electrical contact layer and a second electrical contact layer, and the first electrical contact layer is in contact with the sidewall of the first doped region far from the gate structure; and the second electrical contact layer is located on the first electrical contact layer, and a distance between the second electrical contact layer and the gate structure is greater than a distance between the first electrical contact layer and the gate structure in a direction perpendicular to the sidewall of the gate structure.
3. The semiconductor structure according to claim 2, wherein a top surface of the first electrical contact layer is not higher than the surface of the substrate; and a sidewall of the second electrical contact layer far from the gate structure is flush with a sidewall of the first electrical contact layer far from the gate structure, and a width of the second electrical contact layer is shorter than a width of the first electrical contact layer in the direction perpendicular to the sidewall of the gate structure.
4. The semiconductor structure according to claim 2, wherein the first electrical contact layer is in contact with an entire sidewall of the first doped region far from the gate structure.
5. The semiconductor structure according to claim 2, wherein a length of the second electrical contact layer is shorter than or equal to a length of the first electrical contact layer in an extension direction of a sidewall of the doped region.
6. The semiconductor structure according to claim 2, wherein a material of the first electrical contact layer is the same as a material of the second electrical contact layer.
7. The semiconductor structure according to claim 2, wherein the dielectric layer comprises a first dielectric layer and a second dielectric layer that are sequentially arranged along a direction far from the sidewall of the gate structure; a sidewall of the first dielectric layer far from the gate structure is flush with the sidewall of the first doped region far from the gate structure; and the second dielectric layer is located between the first dielectric layer and the second electrical contact layer.
8. The semiconductor structure according to claim 1, wherein a width of the electrical contact layer remains unchanged in a direction perpendicular to the surface of the substrate.
9. The semiconductor structure according to claim 1, wherein the second doped region and the first doped region are sequentially arranged along a direction far from the sidewall of the gate structure.
10. The semiconductor structure according to claim 1, wherein the first doped region is formed in the second doped region, and a sidewall of the second doped region far from the gate structure is flush with the sidewall of the first doped region far from the gate structure.
11. The semiconductor structure according to claim 1, wherein the electrical contact layer further comprises a metal semiconductor compound layer, a resistivity of a material of the metal semiconductor compound layer is greater than a resistivity of a material of the first doped region, and the metal semiconductor compound layer is in contact with the sidewall of the first doped

region far from the gate structure.
12. A forming method of a semiconductor structure, comprising: providing a substrate and a gate structure, wherein the gate structure is located on the substrate; performing ion implantation on the substrate to form a plurality of doped regions in the substrate, wherein the doped regions are located at two sides of the gate structure; wherein one of the plurality of doped regions comprises a first doped region and a second doped region, a concentration of doped ions in the first doped region is greater than a concentration of doped ions in the second doped region, and the first doped region is far farther from a sidewall of the gate structure than the second doped region; removing part of the substrate, to form a trench, wherein the trench exposes a sidewall of the first doped region far from the gate structure; forming an electrical contact layer, wherein the electrical contact layer fills up the trench and is in contact with the sidewall of the first doped region far from the gate structure, and a top surface of the electrical contact layer is higher than a surface of the substrate; and forming a dielectric layer, wherein the dielectric layer fills a space between the electrical contact layer and the gate structure.
13. The forming method of a semiconductor structure according to claim 12, wherein the forming the electrical contact layer and the forming the dielectric layer comprise: forming the dielectric layer on a surface of the gate structure after the gate structure is formed, wherein a sidewall of the dielectric layer far from the gate structure is flush with the sidewall of the first doped region far from the gate structure; and forming the electrical contact layer after the trench is formed, wherein the electrical contact layer fills the trench and is located on the sidewall of the dielectric layer, and a width of the electrical contact layer remains the same in a direction perpendicular to the surface of the substrate.
14. The forming method of a semiconductor structure according to claim 13, wherein the forming the dielectric layer comprises: forming an initial dielectric layer, wherein the initial dielectric layer covers the surface of the gate structure and the surface of the substrate, and a thickness of the initial dielectric layer located on the sidewall of the gate structure is equal to a width of a top surface of the doped region in a direction perpendicular to the sidewall of the gate structure; and etching part of the initial dielectric layer before the substrate is etched, to expose the surface of the substrate, wherein the remaining part of the initial dielectric layer is used as the dielectric layer.
15. The forming method of a semiconductor structure according to claim 12, wherein the forming an electrical contact layer and the forming a dielectric layer comprise: forming a first dielectric layer on a surface of the gate structure after the gate structure is formed, wherein a sidewall of the first dielectric layer far from the gate structure is flush with the sidewall of the first doped region far from the gate structure; forming an initial electrical contact layer after the trench is formed, wherein the initial electrical contact layer fills up the trench, and the initial electrical contact layer is located on one side of the gate structure; removing part of the initial electrical contact layer located on the sidewall of the gate structure, wherein the remaining part of the initial electrical contact layer is used as the electrical contact layer, and a sidewall of the electrical contact layer and the sidewall of the gate structure enclose a groove; and filling the groove, to form a second dielectric layer, wherein the first dielectric layer and the second dielectric layer constitute the dielectric layer.